

Street-Level Competitive Ride-Hailing with Fixed-Rival Defensive PSRO

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Abstract

Competitive ride-hailing has typically been modeled at the zone level, where fleet decisions are optimized over coarse spatial aggregates. We study whether moving to street-level, per-vehicle competition changes the practical learning problem enough to justify a different modeling stack. We introduce a SUMO-backed street-level competitive ride-hailing framework calibrated with NYC TLC trip records, then evaluate three policy layers: a fixed-rule greedy baseline, a feature-engineered street-level controller, and exploratory U-Net / hybrid residual extensions that test whether learned local corrections can improve the structured controller. We also propose Fixed-Rival Defensive Policy-Space Response Oracles (FRDF-PSRO), a fixed-rival PSRO-style protocol that reweights opponent mixtures toward strategically difficult rival variants. In Manhattan-scale simulations, the street-level feature-engineered controller outperforms the tested zone-level baselines and improves over the tested fixed-price greedy baseline, while FRDF-PSRO improves the selected fixed-rival robustness objective relative to fixed-rival standard PSRO. These robustness gains are scenario-dependent and do not imply universal dominance in direct comparison against the strongest standard-PSRO anchor. The residual study is correspondingly exploratory: some small gains are possible, but they are sensitive to architecture and optimization regime. Overall, the results support street-level competitive simulation and structured street-level control as useful tools for studying strategic ride-hailing behavior.

1 Introduction

Autonomous ride-hailing shifts competition from indirect platform incentives to direct fleet control: each company decides where each vehicle goes, what price it quotes, and which travelers it serves. Most prior learning-based work studies this problem at the **zone level**, where agents observe and act on coarse aggregates (Toft et al. 2026; Chen et al. 2024; Xie, Liu, and Chen 2023). We study whether street-level competition changes outcomes materially enough that a per-vehicle formulation is worth the added complexity.

This resolution change is not only about finer maps. At street level, strategic interactions become strongly local and often non-transitive: policies that work in one neighborhood configuration can fail in another, and a competitor can exploit specific blocks, corridors, or local demand pockets that are invisible after zone aggregation (Lanctot et al. 2017). Our

experiments therefore test street-level control as a substantively different design point rather than assuming that it is only a higher-resolution zone model.

These considerations motivate a modeling stack with three components: a SUMO-backed microscopic competitive ride-hailing environment calibrated on NYC TLC and OpenStreetMap data (Lopez et al. 2018; NYC Taxi and Limousine Commission 2026; OpenStreetMap contributors 2026); a street-level policy family centered on a stable feature-engineered controller with optional U-Net residual refinement; and a fixed-rival robustness protocol in which FRDF-PSRO defensively reweights a payoff-derived mixture over normal and adversarial rivals.

Our contributions are fourfold:

- **Street-level competitive simulation framework.** We develop a microscopic, SUMO-backed ride-hailing environment with per-vehicle control, multiple competing AV fleets, mixed human-driven traffic, and NYC TLC-calibrated demand. To our knowledge, this is the first framework in our cited comparison set to combine competitive ride-hailing, per-vehicle street-level control, and microscopic SUMO traffic, bridging the gap between microscopic but non-competitive simulators (Guériau and Dusparic 2018; Guériau et al. 2020) and competitive but zone-level simulators or multi-provider MoD studies (Toft et al. 2026; Andreasen et al. 2025; Engelhardt et al. 2022; Wang, de Almeida Correia, and Lin 2022).
- **Hybrid residual policy design space.** We implement and document a policy family that decomposes decision making into a stable feature-engineered backbone and an expressive U-Net-style residual (Ronneberger, Fischer, and Brox 2015). The residual branch is exposed as a configurable family of exploration and training variants: latent perturbation, discrete strategy mode, and rollout-level parameter noise. This family provides a controlled testbed for studying whether learned local corrections can improve a strong structured street-level baseline.
- **Fixed-rival Defensive PSRO protocol.** We formalize defensive reweighting over an opponent set and instantiate it in a fixed-rival robustness protocol. Although the rival side does not co-evolve, the protocol still preserves core PSRO structure: an empirical payoff matrix is maintained, a meta-strategy is solved from that game, and new

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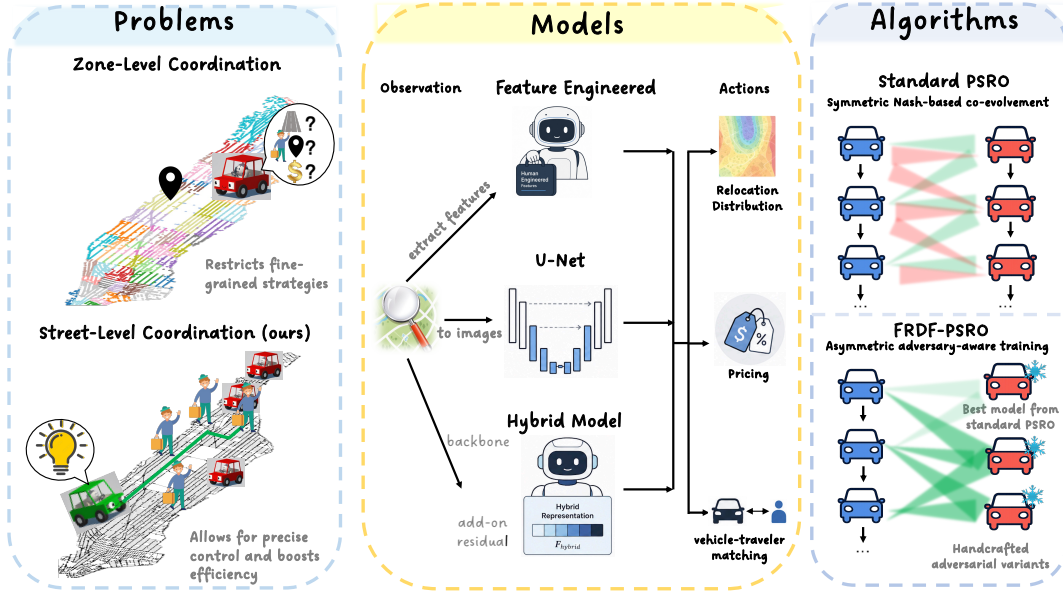


Figure 1: Framework overview. The figure contrasts zone-level coordination with our street-level setting, then places the main policy families and training protocols in one pipeline: feature-engineered street-level control, U-Net and hybrid residual extensions, standard PSRO, and FRDF-PSRO with adversarial rival variants.

ego best responses are trained against the resulting rival mixture. The fixed-rival design is not merely a simplification; it allows us to train against off-Nash adversarial rival variants that would not arise naturally from ordinary co-evolution alone.

- **Empirical validation.** Using full-Manhattan SUMO simulations with NYC TLC-calibrated GMM demand, we study (a) street-level versus zone-level modeling, (b) learned street-level control versus a fixed-rule greedy baseline, (c) FRDF-PSRO versus fixed-rival standard PSRO under normal and adversarial rival variants, including 20% price undercutting, two-fold rival fleet size, and 25% rival brand-prior advantage, and (d) an exploratory U-Net/Hybrid sensitivity suite whose purpose is to test whether further improvement over the feature-engineered controller is stable under conservative, intermediate, and aggressive training regimes.

2 Related Work

Our work connects four threads: competitive ride-hailing, microscopic traffic simulation, population-based game solving, and robust RL. The closest competitive AMoD learning works study pricing plus rebalancing under duopoly competition, but they remain **zone-level**, use **per-operator** agents, and treat robustness largely as in-distribution convergence rather than strategic defense against off-equilibrium opponents (Chen et al. 2024; Andreasen et al. 2025; Toft et al. 2026). Complementary multi-provider simulator studies also model operator competition and cooperation, but without our street-level per-vehicle PSRO-style training stack (Engelhardt et al. 2022; Wang, de Almeida Correia, and Lin 2022). By contrast, our framework operates at **street level**

with **per-vehicle** control, models mixed-autonomy traffic in a microscopic SUMO environment, and treats adversarial rival behavior as a first-class training concern.

On the simulation side, prior SUMO-based ride-hailing systems largely focus on single-operator or non-competitive settings (Guériau and Dussapic 2018; Guériau et al. 2020; Akman et al. 2025; Gammelli et al. 2022). On the game-solving side, PSRO and its descendants provide the natural population-based foundation for non-transitive competition (Lanctot et al. 2017; Marris et al. 2021; Muller et al. 2020; McAleer et al. 2020, 2022; Liu et al. 2021; Li et al. 2024; Liu et al. 2026; Bighashdel, Simão, and Oliehoek 2026; Lian et al. 2025), but they do not directly target strategic robustness to explicit off-equilibrium rival variants. Robust RL and DRO methods largely focus on *environmental* uncertainty rather than strategically adverse opponents (Iyengar 2005; Nilim and El Ghaoui 2005; Smirnova, Dohmatob, and Mary 2019; Ben-Tal et al. 2013; Duchi and Namkoong 2021; Chow et al. 2015; Tamar, Glassner, and Mannor 2015; Qu et al. 2026; Pei et al. 2025; Gonzales et al. 2026). We therefore combine street-level microscopic competition with a fixed-rival defensive PSRO protocol aimed specifically at strategic robustness. A fuller comparison table is provided in the supplementary document.

3 Problem Formulation

We model competitive ride-hailing with company-operated CAV fleets as a partially observable Markov game $\langle \mathcal{C}, \mathcal{S}, \{\mathcal{O}_c\}, \{\mathcal{A}_c\}, P, \{r_c\}, H \rangle$, where companies $c \in \mathcal{C}$ control fleets over a finite horizon H . The key departure from prior work is that both observation and action are **street-level and per vehicle**. Each vehicle observes its own state, neigh-

boring same-company vehicles, rival AVs, nearby travelers, and time context, while mixed-autonomy background traffic is represented through the SUMO state and induced traffic dynamics rather than through a separate HDV observation channel in the current policy input. Each vehicle then outputs a relocation destination, a quoted price, and traveler-specific willingness scores. Competition is resolved at the **individual traveler level**: travelers first decide whether to ride as a function of the best available price, then choose a company based on a distance-weighted effective price, after which within-company matching uses willingness scores to assign vehicles. Company reward is total completed-trip revenue. The supplementary document records the implementation-specific formulas used in the experiments.

4 Method

Our method addresses three challenges: (1) the need for a policy architecture that is stable yet expressive at street level, (2) the need for population-based training in a non-transitive competitive setting, and (3) the need for robustness to off-equilibrium opponent behaviors. We summarize the main design here and move only implementation-specific details to the supplementary document.

4.1 Microscopic Simulation Infrastructure

The simulator combines full-Manhattan SUMO traffic dynamics with NYC TLC-calibrated day-specific dual GMM demand models and an experiment orchestration layer for PSRO training and rollout management. Concretely, for each day we fit separate pickup and dropoff Gaussian mixture models in (x, y, t) space from TLC trip records, then at simulation time condition the corresponding day-specific mixture on the current epoch time and sample traveler origins and destinations from the resulting spatial distributions. Figure 2 shows representative pickup-demand intensity slices for weekday/weekend and day/night regimes. The current Manhattan demand-generation path fits each focal day separately, so the reported experiments use day-specific rather than cross-day-smoothed GMMs. Additional simulator details are provided in the supplementary document.

4.2 Hybrid Residual Policy Architecture

The model family combines a feature-engineered macro-controller with an optional learned U-Net-style residual branch. The feature-engineered controller supplies stable pricing, willingness, and relocation priors from structured spatial features. Concretely, the FeatEng controller is a compact 21-coefficient policy: 3 relocation coefficients define a discretized grid-level relocation density through traveler attraction plus vehicle and center repulsion; 9 pricing coefficients control bias and distance-decayed responses to rival density, traveler density, and city-center attractiveness; and 9 willingness coefficients control pickup proximity, trip profitability, local congestion, and destination centrality. The residual branch then operates on rasterized local state to provide fine local corrections. In the Hybrid setting, the FeatEng controller is frozen and the residual branch is trained using

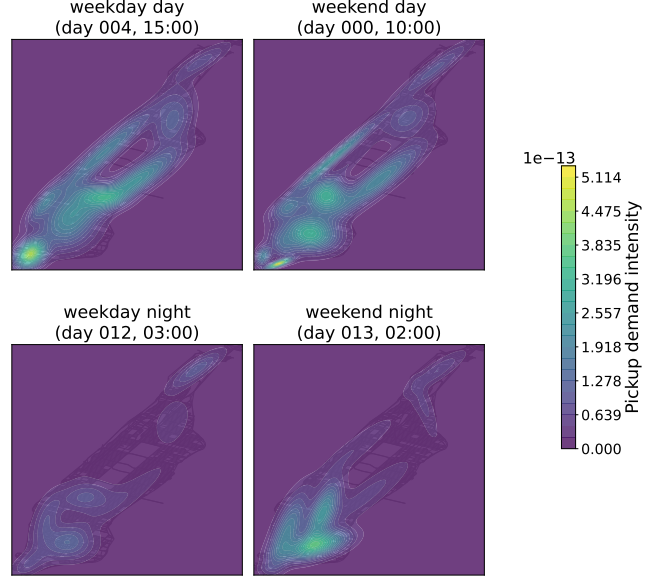


Figure 2: Day-specific TLC-calibrated pickup-demand intensity slices for weekday day, weekend day, weekday night, and weekend night. The stronger daytime panels and distinct weekend-night structure motivate time-conditioned street-level demand.

a revenue-weighted directional objective with regularization toward the backbone.

The codebase currently supports three residual-training variants. All three share the same small U-Net-style scaffold: an 8-channel raster observation, plus a learned spatial embedding and a projected latent or mode map concatenated before the first convolution, followed by a two-stage encoder, bottleneck, and skip-connected decoder with separate relocation, price, and willingness heads. The three rollout-consistent variants are **latent perturbation**, which samples one continuous latent vector per rollout; **strategy mode**, which samples one discrete strategy id per rollout and centers outputs against the zero-mode path; and **weight noise**, which applies one parameter-space perturbation to the residual branch per rollout. Full architectural details and implementation notes are provided in the supplementary document.

4.3 Fixed-Rival Defensive PSRO (FRDF-PSRO)

Standard PSRO trains against the payoff-derived opponent mixture. Our paper-facing robustness protocol instead fixes a pool of strong nominal and adversarial rival variants, maintains a payoff matrix against that fixed pool, solves a rival meta-strategy from that empirical game, and trains only the ego population. FRDF-PSRO then *reweights* the payoff-derived rival mixture toward variants that are harder for the strongest incumbent. With rival meta mixture σ and strongest-incumbent payoffs R_{bj} , the defensive distribution is

$$q_j(\tau) \propto \sigma_j \exp(-\tau \tilde{R}_{bj}), \quad \tilde{R}_{bj} = \frac{R_{bj} - \bar{R}_b}{s_{R_b}}, \quad (1)$$

where $\tau = 0$ recovers fixed-rival standard PSRO and larger finite τ emphasizes more difficult rival variants. In the current implementation, this empirical game is solved as a *zero-sum surrogate robust-ego game*: the ego payoff is revenue and the rival payoff is set to its negation, so the solved meta-strategy should be interpreted as a robustness-oriented scenario mixture rather than as a measured general-sum rival utility model. This design preserves the core PSRO structure—population expansion, payoff-matrix updates, and meta-strategy computation—while allowing explicit off-Nash adversarial variants such as price undercutting, fleet expansion, and brand bias. The supplementary document provides extended algorithmic details and implementation notes.

5 Experiments

We design experiments to answer four questions:

1. **Does street-level modeling yield measurable advantages over zone-level abstraction?** (Section 5.2)
2. **Does the learned street-level feature-engineered controller justify itself relative to a fixed-rule greedy baseline?** (Section 5.2)
3. **Can U-Net or Hybrid extensions reliably improve on the feature-engineered controller, and how sensitive are those gains to architecture and hyperparameters?** (Section 5.3)
4. **Does FRDF-PSRO improve adaptation to fixed adversarial rival variants without sacrificing normal-case performance?** (Section 5.4)

5.1 Experimental Setup

Simulator and demand. All experiments use the full Manhattan SUMO road network with $K = 2$ competing companies and, unless stated otherwise, $N = 20$ autonomous vehicles per company. Demand comes from the day-specific TLC-calibrated dual GMM described in Section 4. Validation uses 5-minute decision epochs and averages over four temporal slices: weekday day, weekend day, weekday night, and weekend night. Unless otherwise noted, the simulator runs at the default demand intensity and the same temporal evaluation suite is reused across direct comparisons.

Training. The street-vs-zone baselines use two-sided PSRO. Feature-engineered and hybrid runs warm-start each best response from the current strongest incumbent under the solved meta-strategy. The fixed-rival robustness experiments use a reduced protocol with three ego-expansion iterations, two best-response iterations per expansion, 120 optimizer steps per best response, eight rollout workers, learning rate 0.01, entropy regularization 0.001, and Adam.

Evaluation. We report the **mean revenue per company per evaluation window**. The street-vs-zone and learned-versus-greedy comparisons use direct head-to-head evaluation over the four temporal slices, with the greedy baseline using a fixed unit-distance price and simple hand-coded relocation / willingness rules. The fixed-rival robustness study selects the best checkpoint under the fixed-rival robustness objective and reports four-seed means; direct comparison

against the standard-PSRO FeatEng anchor is treated as diagnostic rather than primary.

5.2 Street-Level vs. Zone-Level Modeling

Motivation. If street-level fidelity is merely a precision upgrade, it should not produce materially different outcomes from zone-level abstraction beyond smoother estimates. We instead test whether fine-grained spatial information enables **different strategic behaviors**, such as positioning vehicles at specific intersections, reacting to rival proximity at the individual-vehicle level, and selectively serving nearby travelers rather than broadcasting uniform availability.

Setup. We compare street-level agents against a ladder of zone-level baselines that differ in how much spatial structure they preserve:

- **Zone-level agents:** We use the real Manhattan TLC taxi zones already employed in the demand pipeline, rather than an artificial square grid. Each vehicle observes a zone-level summary centered on its current zone, including same-company vehicle counts, rival vehicle counts, traveler counts, matched-traveler counts, mean pickable price, centroid distance, time-of-day, and related derived features. The agent still emits per-vehicle actions through the existing simulator interface, but relocation is scored at the zone level and then mapped back to valid grid-level relocation probabilities within each target zone. We evaluate three zone variants under the same PSRO framework: **Zone-Independent**, an adjacency-unaware per-zone scorer using only each zone’s own features; **Zone-GraphLinear**, an adjacency-aware linear scorer using local, one-hop, and two-hop zone feature aggregation; and **Zone-GraphNonlinear**, an adjacency-aware nonlinear scorer over the same aggregated zone features. We then define **Zone-Best** as the best-performing member of this ladder under a fixed tuning and validation budget.
- **Street-level agent:** Our full street-level model as described in Section 3, with per-vehicle observations and actions on the Manhattan SUMO network.

All variants are trained under the same PSRO protocol and rollout budget. The zone ladder is first compared internally to determine **Zone-Best**. We then compare **Zone-Best** against **street-level FeatEng**, which is the central street-versus-zone resolution ablation in this paper. We also compare **street-level FeatEng** against a **fixed-rule greedy baseline** that reuses simple hand-coded relocation and willingness heuristics while keeping a single fixed unit-distance price throughout evaluation. This comparison therefore tests adaptive structured control against a simpler fixed-price rule stack, rather than claiming exact price matching.

Experiment design. This study is intentionally staged. First, we run an internal zone-ladder ablation comparing **Zone-Independent**, **Zone-GraphLinear**, and **Zone-GraphNonlinear**. This isolates whether gains come from adjacency awareness, from nonlinearity, or neither. Second, we promote the strongest of the three to **Zone-Best** and compare it against **street-level FeatEng**. This is the most interpretable resolution comparison because both sides are

relatively structured. All variants are evaluated on revenue per epoch, trips completed, average quoted price, pickup distance or wait proxy, relocation efficiency, and compute cost.

Results. In the ordered tournament, the street-level FeatEng model beats all three trained zone-level variants in direct head-to-head play. The strongest zone baseline by overall tournament revenue is Zone-Independent. Against this Zone-Best model, street-level FeatEng earns 580.6 mean ego revenue versus 468.8 for the zone rival, a 23.9% direct revenue advantage. Across all ordered tournament appearances, street-level FeatEng averages 564.0 revenue, while the three zone-level models average 401.3 revenue, a 40.5% advantage. Against the fixed-rule greedy baseline, role-swapped head-to-head validation yields a role-balanced FeatEng average revenue of 633.3 versus 598.5 for the greedy baseline, a 5.8% gain. This supports the claim that learned street-level adaptation is adding value beyond a hand-specified fixed-price controller.

Findings.

- The street-level model wins the direct matchup against every zone-level variant, with revenue advantages of 23.9%, 24.1%, and 21.0% when used as the ego model.
- The largest measured difference is not trip count in the Zone-Best comparison, but trip quality and positioning: street-level control achieves 25.0% higher average trip revenue and 63.5% shorter pickup distance.
- Relative to the fixed-rule greedy baseline, the learned street-level FeatEng controller improves role-balanced head-to-head revenue by 5.8%, indicating that the gain is not only a consequence of moving from zone abstraction to street resolution, but also of learning adaptive street-level behavior.
- The result supports the claim that, in this tested controller family, street-level fidelity changes the policy’s actionable information rather than merely refining a zone-level estimate; it should not be read as a pure resolution-only ablation with every other modeling choice held fixed.

5.3 Hybrid Policy Architecture Ablation

Setup. We compare three architectures under the same simulator conditions: FeatEng alone, U-Net alone, and Hybrid with a frozen FeatEng backbone plus one of three residual-training variants (latent perturbation, strategy mode, or weight noise). For each tiny residual variant we test conservative, intermediate, and aggressive hyperparameter regimes.

Results. The hybrid study is best read as a sensitivity analysis around a strong FeatEng baseline. Smoke-stage behavior already shows instability under aggressive settings: latent perturbation moves from near-neutral to strongly negative, strategy mode stays weak throughout, and aggressive ES / weight-noise inflates the smoke metric while also increasing rival revenue. Direct head-to-head validation against FeatEng is therefore the decisive test. The best direct result is **latent perturbation / intermediate**, with role-balanced mean revenue 639.4 for Hybrid versus 634.6 for FeatEng (+0.77%). Latent perturbation / conservative is only slightly positive (+0.32%), latent perturbation / aggressive turns negative

(−0.91%), strategy mode stays near identity-like behavior at about +0.50%, and all three ES / weight-noise candidates are negative under direct validation, with the aggressive setting collapsing badly (−36.9%). The practical conclusion is modest: learned residuals can produce small gains, but those gains are variant-dependent and substantially less robust than the street-versus-zone and FRDF-PSRO results.

Summary.

- **FeatEng only:** This is the stable street-level baseline used for the completed resolution and fixed-rival robustness results.
- **U-Net only:** This setting is useful mainly as an ablation on what is lost when the structured backbone is removed.
- **Hybrid:** The implemented residual-training variants are latent perturbation, strategy mode, and weight noise; the most competitive direct results come from latent perturbation, while aggressive ES is clearly unstable.

5.4 Robustness to Adversarial Competition

Setup. We compare two fixed-rival oracle variants: fixed-rival standard PSRO ($\tau = 0$) and FRDF-PSRO ($\tau > 0$). A pilot sweep over $\tau \in \{0.5, 1.5, 3.0\}$ selected $\tau = 1.5$ by best-checkpoint mixture revenue (641.3 versus 640.9 for $\tau = 3.0$). The fixed rival pool contains seven normal copies of the best feature-engineered rival plus three adversarial variants: 20% price undercutting, fleet asymmetry, and 25% brand-prior bias. This preserves a payoff matrix and solved rival meta-strategy while keeping the rival pool fixed, which is useful because these off-Nash adversarial variants are introduced deliberately rather than through co-evolution. We report the stronger fleet-2x stressor because the milder fleet perturbation did not reliably reduce ego revenue across seeds. Selected best checkpoints are then evaluated under four scenarios: normal, undercutting, fleet asymmetry, and brand bias, without validation-time reweighting.

Results. Table 2 reports mean ego revenue across four seed bases under the validation-selected fixed-rival robustness objective. On this objective, FRDF-PSRO improves the mean over fixed-rival standard PSRO in the reported normal setting and in each reported adversarial category. The uniform mixture over the four validation scenarios improves from 609.9 to 634.4 mean ego revenue, a 4.0% gain. This is the main robustness claim of the paper: defensive reweighting improves the selected fixed-rival robustness objective relative to the non-defensive fixed-rival baseline. It should not be read as a universal direct-head-to-head superiority claim against the standard-PSRO FeatEng anchor, nor as a claim about fully co-evolving PSRO. The aggregate result is also not a point-wise dominance claim at the seed level: FRDF-PSRO wins all four normal validations, but wins 2/4 under price undercutting, 3/4 under two-fold fleet asymmetry, and 3/4 under brand-prior bias.

As a diagnostic cross-check, Table 3 compares three concrete models directly on a shared stronger head-to-head suite: the standard-PSRO FeatEng anchor, the best fixed-rival standard checkpoint, and the best FRDF-PSRO checkpoint. This comparison answers a different question from Table 2. It asks whether the robustness-trained models beat the strongest

Table 1: Street-level vs. zone-level: final comparison using the strongest zone baseline in the ordered tournament.

Metric	Zone-Best	Street-level	Δ (%)
Mean revenue / eval. window / company	468.8	580.6	+23.9%
Trips completed / eval. window / company	58	58	0.0%
Avg. trip revenue	7.48	9.35	+25.0%
Avg. pickup distance (m)	5591.9	2041.6	-63.5%

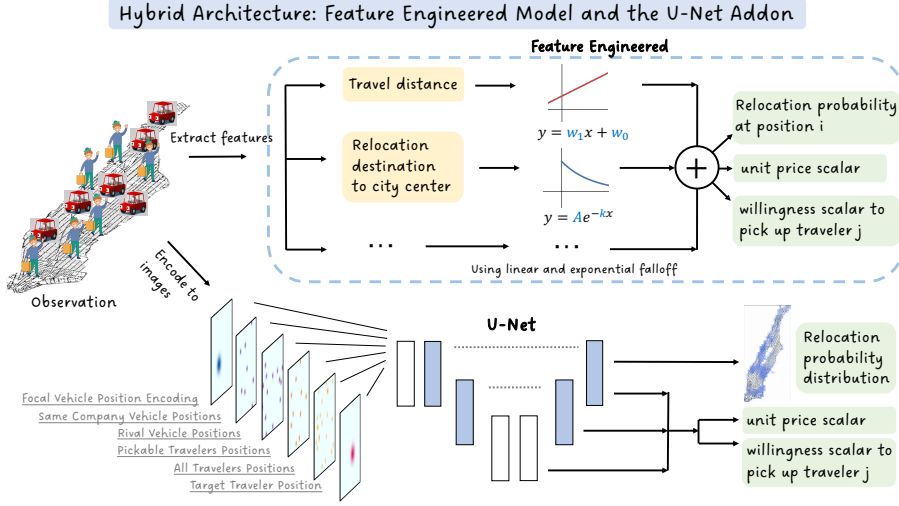


Figure 3: Comparison among representative street-level learned policy families. The figure contrasts the feature-engineered model with the best-performing tiny U-Net addon variant, emphasizing the trade-off between structured inductive bias and learned local refinement.

Table 2: FeatEng fixed-rival robustness comparison across four seed bases, using the best checkpoint from each run under the fixed-rival robustness objective. FRDF-PSRO uses the defensive-temperature setting selected by a pilot sweep over $\tau \in \{0.5, 1.5, 3.0\}$, which chose $\tau = 1.5$ by best-checkpoint mixture revenue. Values are mean ego revenue. Percentages are relative to fixed-rival standard PSRO.

Scenario	Fixed-rival standard PSRO	FRDF-PSRO
Normal	623.7	650.0 (+4.2%)
Undercut-20	618.1	634.6 (+2.7%)
Fleet-2x	617.4	646.1 (+4.7%)
Brand-prior-plus-25	580.4	606.9 (+4.6%)
Uniform mixture	609.9	634.4 (+4.0%)

standard-PSRO anchor in direct ordered-pair play, rather than whether FRDF-PSRO improves the fixed-rival robustness objective over the non-defensive fixed-rival baseline.

Interpretation.

- **Objective-level improvement:** FRDF-PSRO improves mean revenue on the reported fixed-rival robustness objective, both in the four-scenario aggregate and in the mean of each reported scenario category.
- **Direct-comparison caveat:** On the stronger shared direct-validation suite, the standard-PSRO FeatEng anchor remains strongest in three of the four reported scenarios, while the fixed-rival models show only a small advantage under severe price undercutting.

- **Seed-level variability:** The result is not uniform at the seed level; price undercutting remains the weakest case, with FRDF-PSRO winning 2/4 seeds.
- **Fleet stressor validity:** Under the two-fold fleet stressor, fixed-rival standard PSRO’s ego revenue falls below its normal-setting revenue in all four seeds, confirming that this variant creates reliable adversarial pressure.

6 Conclusion

This paper presents a street-level framework for competitive ride-hailing with company-operated autonomous vehicle fleets. The framework combines a SUMO-backed microscopic competitive ride-hailing environment on the full

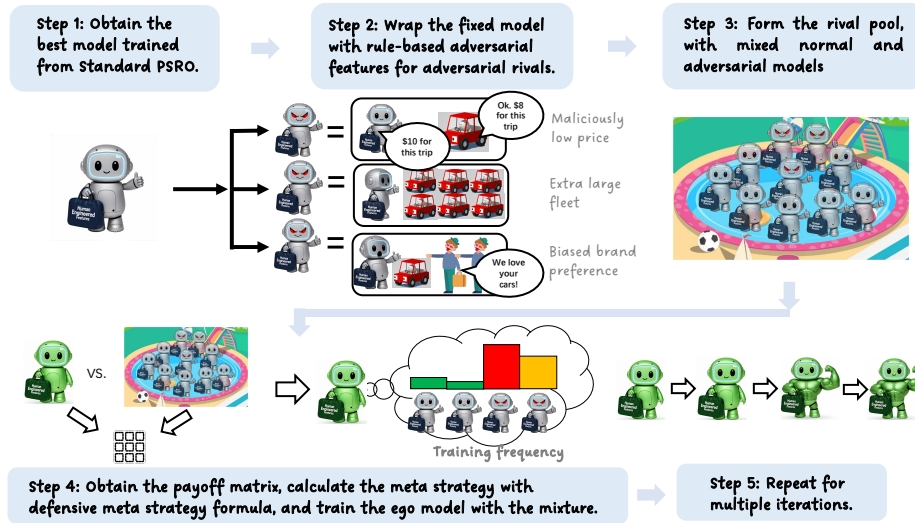


Figure 4: FRDF-PSRO pipeline. Starting from the best model obtained by standard PSRO, we construct fixed adversarial rival variants such as price undercutting, fleet enlargement, and brand-preference bias, form a mixed rival pool, compute the empirical payoff matrix and defensive meta-strategy, and train ego best responses against that reweighted rival mixture over multiple iterations.

Table 3: Direct three-way comparison on the shared stronger validation suite. Values are role-balanced mean ego revenue from ordered head-to-head play under the same opponent scenario. The ‘standard-PSRO FeatEng anchor’ is the best saved FeatEng model from the standard two-sided PSRO pipeline; the other two entries are the selected best checkpoints from fixed-rival standard PSRO and FRDF-PSRO. This table is diagnostic rather than the primary robustness objective.

Scenario	Standard-PSRO FeatEng anchor	Fixed-rival standard best	FRDF-PSRO best
Normal	<u>672.3</u>	659.8	643.9
Undercut-50	<u>471.3</u>	<u>474.8</u>	473.3
Fleet-plus-50	<u>774.1</u>	743.9	715.5
Brand-bias	<u>639.3</u>	628.9	615.1

Manhattan road network, a strong feature-engineered street-level controller, a fixed-rule greedy baseline used to calibrate the value of learned adaptation, a family of U-Net and hybrid residual extensions, and an FRDF-PSRO training protocol that reweights rival mixtures toward strategically difficult variants.

Empirically, the results show that the tested street-level controller stack yields higher revenue than the tested zone-level abstractions, that the learned FeatEng street-level controller outperforms the fixed-rule greedy baseline by 5.8%, and that FRDF-PSRO improves aggregate adaptation to adversarial rival variants relative to fixed-rival standard PSRO in a qualified, scenario-dependent way rather than as a universal per-seed improvement. The hybrid study shows that residual learning can produce modest gains, but those gains are sensitive to variant choice and optimization regime. These conclusions should be read with the paper’s scope limits in mind: the reported study uses a fixed Manhattan simulator configuration, limited fleet sizes, and a fixed-rival robustness protocol rather than fully co-evolving adversaries. It also does not yet include full formal significance testing for the reported small performance deltas. Overall, the results sup-

port street-level competitive simulation and structured street-level control as useful tools for studying strategic ride-hailing behavior. Additional derivations, implementation details, extended related work, and deferred analyses are collected in the supplementary document.

Responsible use. The framework is intended for simulation-based analysis rather than direct deployment. The reported objectives optimize operator revenue, not passenger welfare or fairness, and the adversarial variants are synthetic stress tests rather than pricing recommendations. The TLC-based demand pipeline uses public trip records only to fit aggregate demand models, and the street network is derived from OpenStreetMap under its standard open-data attribution terms.

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